LLM Loan Underwriting Design Guide

Version 4 | USD units | Multi-agent Python implementation handoff

Status: Draft for development and policy validation. No authorization for production automated credit decisions.

1. Currency and amount conventions

• **All monetary amounts are USD (United States dollars).** Every financial field and threshold must carry currency = USD, and all monetary comparisons must use normalized USD amounts.

• Use ISO 4217 **USD**; parse values as Decimal, not binary floating point. Persist amounts as exact USD decimal values (for example, 100000.00).

• USD 100M = USD 100,000,000; USD 250M = USD 250,000,000; USD 25M = USD 25,000,000; USD 100K = USD 100,000.

• If a source value lacks a known currency or unit scale, mark it UNKNOWN; do not silently convert, scale, or treat it as USD.

• Keep distinct **loan_requested_amount_usd**, **tier1_group_credit_usd**, **tier2_group_credit_usd**, **total_group_credit_usd**, and **credit_limit_usd**. Do not conflate these amounts.

2. Source data model and joining rules

• Account -> Loan 1:N; Account <-> Loan M:N through LLC_BI_LEGAL_ENTITIES__c EntityInvolvement with borrower/guarantor role. Validate borrower account, loan ID and role explicitly.

• Account -> nFUSE CreditReport 1:N; select the appropriate dated report by account and role. Loan -> LoanExtension 0..1; Loan -> SCOC_LCRESPONSE 1:N.

• Account -> SCOC_LCRENTITY 1:N; LCRESPONSE -> LCRENTITY 1:N; LCRENTITY -> SCOC_LCRDATASOURCEANALYSIS 1:N; analysis -> SCOC_LCINDICATOR / SCOC_LCCHARACTERISTIC 1:N.

• Query 1 is underwriting master; Query 2 is indicators; Query 4 is credit bureau. Never join on row number or confuse indicator IDs with credit report IDs. Preserve provenance: file, sheet, row, record IDs.

3. USD field mapping

Normalized field	Source / notes
loan_requested_amount_usd	LLC_BI_Amount__c (Loan)
tier1_group_credit_usd	Group_Credit_Amount__c; confirm Tier I scope
estimated_annual_revenue_usd	Estimated_Annual_Revenue__c
business_net_income_usd	Business_Net_Income__c
ebitda_usd	Derive only from approved inputs; EBIT is not automatically EBITDA
cash_and_equiv_usd	SBSS7_cash_and_equiv__c
net_worth_usd / total_assets_usd	SBSS7_net_worth__c / SBSS7_total_assets__c
total_liabilities_usd	SBSS7_total_liabilities__c
total_past_due_usd	nFUSE_Total_Past_Due__c
available_credit_usd	nFUSE_Available_Credit__c
bankruptcy_amount_usd	nFUSE_Bankruptcy_Amount__c

4. Underwriting policy and unresolved conflicts

• Quick App notes: new money <= USD 100,000,000 and Tier I Group Credit <= USD 250,000,000; nonprofits require full underwriting. Express BALOC <= USD 100,000,000 unsecured; Conventional BALOC > USD 100,000,000 secured. Startup BALOC cap USD 25,000,000; investment real estate BALOC cap USD 25,000,000.

• **Conflict A:** Quick App strategy image shows requested amount USD 100M, but knock-out text says USD 100,000.00. Treat as two unverified interpretations, not interchangeable; do not auto-approve affected cases.

• **Conflict B:** BALOC increase restriction appears as 6 months in product notes and 12 months in decline reasons. Require policy owner resolution.

• Knock-outs shown: negative EBITDA; bankruptcy discharge within 5 years; applicant under 18; average personal FICO < 650; SBSS < 160. Do not substitute business credit score for personal FICO.

• SBSS bands: 160-189 ownership/FICO/amount branch; 190-199 SBA path; >=200 ownership and NAICS path. High-risk NAICS examples: 23, 4451, 45, 8111, 8121, 8123. These are candidate paths, not guaranteed approvals.

• OFAC alert and credit freeze require designated compliance/manual workflow; do not let LLM autonomously deny based on an unverified screening alert. Unsatisfied judgments, collections, and past due require confirmed policy definitions and evidence.

5. Multi-agent Python responsibilities

• IngestionAgent: read all three workbooks; validate schema, units and source provenance; report missing required fields.

• EntityResolutionAgent: build loan-centric account/loan/role/credit-report/SBSS/analysis relationships; fail closed on ambiguous joins.

• FeatureAgent: normalize USD Decimals; compute approved features; distinguish unknown from zero; capture field-level evidence.

• PolicyAgent (deterministic Python, not LLM): evaluate versioned eligibility, knock-outs and SBSS decision branches; output PASS/FAIL/UNKNOWN and policy references.

• ExplanationAgent (LLM): summarize fixed rule trace and evidence; do not choose, override, or invent a final lending decision.

• ReviewAgent: route unresolved data, policy, compliance, and governance issues to authorized humans; produce audit-ready output.

6. Decision output contract

Return a per-loan JSON record with: mode, policy_version, currency="USD", loan_id, borrower_account_id, source_evidence, normalized_amounts_usd, rule_results, missing_inputs, conflicts, recommendation_path, final_action, explanation, and human_review_required.

TEST recommendation_path: SIMULATED_APPROVE_PATH | SIMULATED_DECLINE_PATH | SIMULATED_REVIEW. Production final_action: MANUAL_REVIEW unless an independently approved decision service / authorized reviewer completes disposition.

Do not report confidence as a calibrated approval probability unless calibrated and validated. Separate policy evaluation confidence from evidence completeness.

7. Required tests and acceptance criteria

• USD boundary tests at 100,000; 25,000,000; 100,000,000; 250,000,000; reject ambiguous 100M / 100K rule mapping.

• SBSS 159/160/189/190/199/200; personal FICO 649/650/759/760; ownership 2 years; bankruptcy discharge 5 years; freeze/OFAC positive/negative/unknown.

• Test multiple loans per account, guarantors, multiple bureau reports, orphan IDs, duplicate indicators, missing dates, mixed units, and invalid currency.

• For each processed loan, emit an explainable rule trace, original row references, USD values, missing-field diagnostics and machine-readable validation report.

8. GitHub Copilot implementation request

Use the accompanying **GitHub_Copilot_Multi_Agent_Update_Instructions.md** as the implementation task. Inspect the existing repository and adapt its actual agents, models, prompts, tools and tests rather than inventing an unrelated app. Do not disable policy sign-off or identity controls to force APPROVE/DENY outputs.